The 28th Chinese Physics Olympiad Finals Problems

I. (15 points)

Fix a smooth vertical semicircular rail at A、B, diameter AB＝$2R$; AB is 60° from vertical. A small smooth ring on it has mass m. A light thin smooth elastic cord of stiffness k passes through it, its ends tied at A、B. Directly below A, at C, the cord is unstretched. Release it from rest at C. At t, it reaches C'; C'O makes angle θ with radius OB. Gravity is g. Find the rail force magnitude for the cases （1）θ＝90°;（2）θ＝30°.

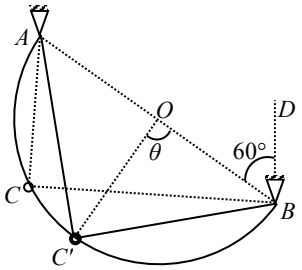


Fig. 28 Final—1

II. (15 points)

A small cart, mass M, length L, is on a level floor. Two springs are fixed near the bottom at both ends, collinear, with rest lengths l_1, l_2 and stiffness k_1, k_2. The other ends meet small balls of masses m_1、m_2, unattached. Initially ball 1 compresses spring 1, holding the system at rest. Ball 2 is locked to the cart floor. Ball 2 is spring 2's rest length from the right end. Release ball 1; when spring 1 regains rest length, unlock ball 2 at once. Balls 1 and 2 collide and stick instantly. From release of ball 1 to spring 2's maximum compression, the cart moves l_3. No friction. Find initial spring 1 length l and spring 2's peak compression Δl_2.

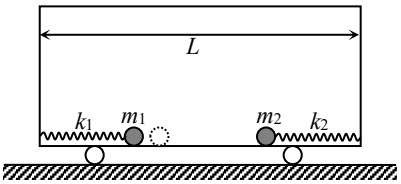


Fig. 28 Final—2

III. (20 points)

Station A circles Earth at radius r_A＝6.73×10^6m. Satellite B circles in the same plane, with radius r_B＝$\frac{3}{2}r_A$. A and B move counterclockwise. Launch a ship （no station recoil） to collect it. For fuel economy, except brief acceleration or braking, it coasts on identical anticlockwise transfer ellipses both ways. Both lie in the orbital plane. Perigee is on the station orbit, apogee on the satellite orbit. Given Earth radius R_e＝6.38×10^6m and surface gravity g＝9.80m/s^2, find:

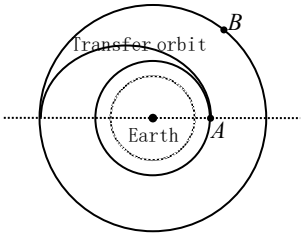


Fig. 28 Final—3

（1） The departure increment Δv_A from station A to the ellipse. If at apogee it meets satellite B, find capture speed increment Δv_A for zero relative motion.

（2） The least time to collect it and return, and the least angle station A must turn around Earth.

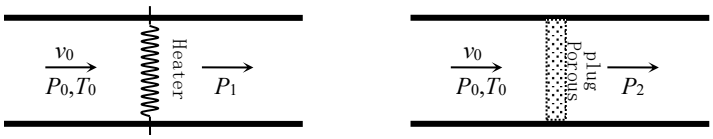
IV. (15 points)

Ideal gas, molar mass μ, flows right in a long straight smooth horizontal adiabatic pipe of area S. Molar internal energy at absolute T is $\frac{5}{2}RT$; R is the gas constant.

（1） A fixed mid-pipe heater supplies constant power W, Fig. （a）. Neglect its resistance to flow. In steady flow the state is nonuniform near the heater, becoming uniform far away. In the uniform steady region on the left, pressure is p_0, temperature T_0, and rightward speed v_0. On the right, steady pressure is p_1. Find that region's temperature T_1.

（2） Use a porous plug instead, Fig. （b）. Left steady temperature and pressure are T_0, p_0, with speed v_0 rightward and right pressure p_2（$p_2<p_0$）.

With no energy exchange of any kind between gas and plug, find the downstream flow speed v_2.



（a） （b）

Fig. 28 Final—4

V. （15 points）

Prism ABC has apex angle α below 90°. A ray in the paper plane hits AB at D at any angle. After one refraction it hits AC, It can reflect totally at AC. The critical angle at AC is θ（$\theta<45°$）, and AC is long enough. Find the range of prism apex α for the following two cases:

（1） The ray enters AB only below the normal;

（2） The ray enters AB only above the normal.

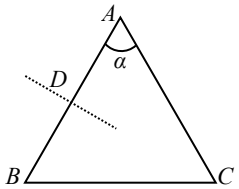


Fig. 28 Final—5

VI. （20 points）

Charge q gives an electric field at distance r of magnitude $E=k_e\frac{q}{r^2}$, with constant k_e. A long straight wire carrying i produces a field at distance r（far below its length ）, of magnitude $B=k_m\frac{2i}{r}$, with constant k_m.

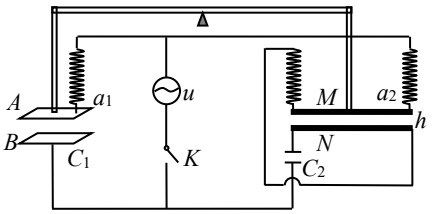


Fig. 28 Final—6

Measure $\sqrt{\frac{k_e}{k_m}}$ in the shown setup using low-frequency （hundreds of hertz） electric and magnetic fields. A、B are horizontal square plates of capacitance C_1. Side a_1（gap far below a_1）. B is fixed; A hangs on one balance hook.

M、N are parallel horizontal thin metal rods: length a_2, gap h（$h\ll a_2$）. Rod N is fixed; M hangs on the other hook. C_2 is a known capacitance; K is a switch. AC voltage u varies with time t as $u=U_0\cos 2\pi ft$; f is frequency. Wire all parts as shown. The balance is level with K open, but loses balance when powered. By

Adjusting AC frequency restores balance （ inertia makes the average AC effect balance it）. Find $\sqrt{\frac{k_e}{k_m}}$. 【Double lines are rigid insulating rods; single lines, wires; curves, flexible massless wires. Ignore wires' magnetic effects on M, N and electric and magnetic fringing】

VII. （20 points）

Two identical light metal springs of stiffness k hang vertically from a horizontal insulating bar, joined to a rigid uniform metal rod: mass m, length l. Length equals their spacing. Uniform field B is inward. Rod, springs and AC source u form a loop, Fig. （a）. Resistance and the loop field relative to B are negligible. In Fig. （b）, an inductor and capacitor are connected in parallel to the same u. If source currents always match, express C, L of （b） using （a）'s known device parameters.

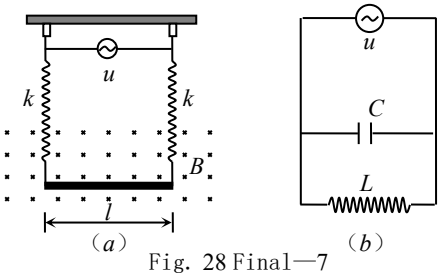


Fig. 28 Final—7

VIII. （20 points）

Steven Chu and two others won the 1997 Nobel in Physics for trapping neutral atoms. These questions explain this trapping. （Note: atomic parameters are averages over many atoms or repeated identical processes of one atom）.

（1） A ground-state atom at rest absorbs photons at v_0 resonantly, reaching its first excited state, Fig. （a）. Atoms move thermally; take speed v_0. A laser meets it head-on. Find its resonance frequency v and the speed change due to absorption. （$hv_0 \ll mc^2$, m is the atomic mass, $h=6.63\times10^{-34}$J•s）.

（2） Absorption is instantaneous, but the excited atom does not return immediately to ground state. Its stay in the excited state is that level's mean lifetime. The first excited state's mean lifetime is τ. On returning to ground state, it at once absorbs another head-on laser photon. Find the mean acceleration between the instants just before successive absorptions. Photons are equally likely in all directions. Very frequent collisions make its mean momentum change zero.

（3） The atom starts at v_0 along positive z; a laser meets it along negative z, causing resonance. At fixed laser frequency, repeated absorption must slow the atom to rest. It crosses a nonuniform field $B(z)$ for trapping, Fig. （c）. Interaction with the field changes the excited-state energy, as in Fig. （b）. At field B, an energy level E becomes $E+\Delta E$, where $\Delta E=\mu B$, μ is a known constant. Find B as a function of z.

（4） Lithium: $m=1.0\times10^{-26}$kg, initial $v_0=1.2\times10^{3}$m • s^{-1}; rest resonance frequency

$\nu_0 = 4.5\times10^{14}$Hz. The first excited state's mean lifetime is $\tau = 5.3\times10^{-8}$s. Using （3）, how long must the magnetic field region be to bring the atom to rest?

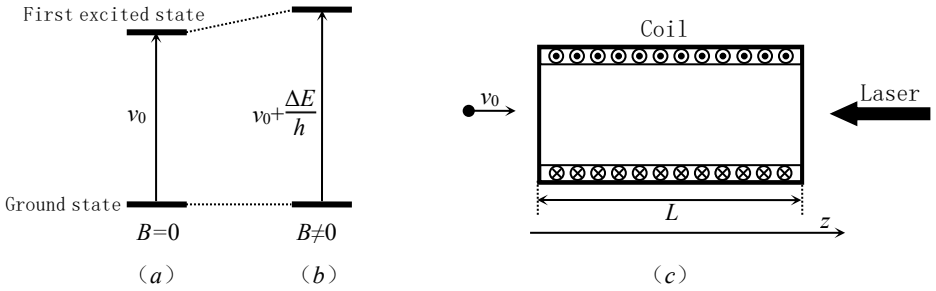


Fig. 28 Final—8